# NETFLIX CONTENT ANALYSIS

*A Data Analysis Project Using Python*

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## Abstract

This project explores a Netflix titles dataset using Python data-analysis libraries. The notebook loads the data into a pandas DataFrame and uses grouping, text splitting, date-time conversion, cross-tabulation, and visualizations to investigate the catalogue's composition and patterns. The questions cover Movies versus TV Shows, content added by year, countries and genres represented, release years, content ratings, movie duration, directors, cast members, the gap between release and Netflix addition, and monthly additions. The dataset contains 8,807 title records in the notebook output. The analysis is exploratory and describes the supplied catalogue snapshot; it does not measure Netflix's complete global catalogue or viewer preferences.

## 1. Introduction

Streaming platforms offer large catalogues of films and episodic programmes. Analysing catalogue metadata can help describe the types of content available, the countries represented, and the timing of releases and additions. This project uses the Netflix titles dataset and Python visual analytics to summarise selected characteristics of the catalogue.

## 2. Objectives

- Compare the number of Movies and TV Shows in the dataset.
- Examine how the number of titles added to Netflix varies by year.
- Identify countries and genres with the highest representation.
- Find release years with the largest number of titles.
- Review the most common content ratings.
- Explore movie duration and its relationship with release year.
- Identify directors and cast members appearing most frequently.
- Examine the time between a title's release year and its Netflix addition year.
- Compare monthly additions and rating/genre composition by content type.

## 3. Dataset Description

The notebook reads a local CSV file named netflix_titles.csv. Its displayed DataFrame has 8,807 rows and fields including show_id, type, title, director, cast, country, date_added, release_year, rating, duration, and listed_in. These columns support the catalogue-level questions in the project. Some metadata fields contain missing values, which the notebook handles selectively —for example, dropping missing country values before country counting and missing cast values before actor counting. The CSV file itself was not included separately in the supplied upload, so this report uses only values and outputs visible in the notebook.

## 4. Tools and Technologies

- Python — programming language used for the analysis.
- pandas — CSV loading, data cleaning, grouping, aggregation, date parsing, and cross-tabulation.
- Matplotlib — charts including line plots, pie charts, bar charts, histograms, and scatter plots.
- Seaborn — imported in the notebook, although the shown analysis primarily uses Matplotlib.
- NumPy — imported for a small numerical array demonstration.

## 5. Methodology

### Data loading

The CSV is loaded with pandas.read_csv() into a DataFrame.

### Date preparation

The date_added field is converted to a datetime value and its year and month are extracted for time-based summaries.

### Categorical analysis

value_counts(), groupby(), and crosstab() are used to count titles by type, rating, country, genre, director, and cast member.

### Multi-value fields

Country, genre, and cast strings are split into individual values and exploded so each listed item can be counted separately.

### Movie duration

Movie rows are selected and the numeric part of the duration field is extracted for duration analysis.

### Visualisation

The notebook creates line charts, pie charts, horizontal and vertical bar charts, histograms, and a scatter plot.

## 6. Analysis and Results

### 6.1 Catalogue composition: Movies vs TV Shows

The notebook reports 6,131 Movies and 2,676 TV Shows, for a total of 8,807 records. That corresponds to approximately 69.6% Movies and 30.4% TV Shows. Movies therefore make up the larger share of the dataset. A pie chart is used to display this comparison.

### 6.2 Titles added by year

The notebook converts date_added to datetime, extracts year_added, and plots yearly counts as a line chart. The printed group-by output for years 2008–2021 shows additions rising substantially from 2015, reaching their highest listed value in 2019 (1,424 Movies and 592 TV Shows), then declining in 2020 and 2021. These values reflect the date_added values present in this dataset, not the total number of titles Netflix added in every market.

### 6.3 Countries represented

The country field is split on commas and exploded so titles associated with multiple countries contribute to each listed country. The notebook then selects the ten most frequent country entries and plots them as a horizontal bar chart. The exact top-ten counts are not printed in the captured notebook output, so this report does not assign numerical values or rank individual countries.

### 6.4 Genre/category representation

The listed_in field is split into individual genre/category labels and counted. The notebook visualises the ten most frequent labels and later compares the number of Movies and TV Shows within the ten most common genres. Exact genre counts are not shown in the captured text output.

### 6.5 Release years with the most titles

The notebook groups records by release_year and prints the ten largest counts. The leading years shown are:

| Release year | Number of titles |
|---|---|
| 2018 | 1,147 |
| 2017 | 1,032 |
| 2019 | 1,030 |
| 2020 | 953 |
| 2016 | 902 |
| 2021 | 592 |
| 2015 | 560 |
| 2014 | 352 |
| 2013 | 288 |
| 2012 | 237 |

Within this dataset, 2018 has the highest recorded count of titles by release year (1,147), followed by 2017 (1,032) and 2019 (1,030). This is distinct from the year a title was added to Netflix.

## 6.6 Content ratings

The ten most common ratings printed by the notebook are:

| Rating | Count |
|---|---|
| TV-MA | 3,207 |
| TV-14 | 2,160 |
| TV-PG | 863 |
| R | 799 |
| PG-13 | 490 |
| TV-Y7 | 334 |
| TV-Y | 307 |
| PG | 287 |
| TV-G | 220 |
| NR | 80 |

TV-MA is the most frequent rating in the printed output, followed by TV-14. Ratings combine different classification systems for television and film, so they should be interpreted as catalogue metadata rather than as a single uniform scale.

## 6.7 Movie duration

The notebook selects Movie records, extracts the numeric portion of duration, and prepares a histogram. The chart is intended to show how movie runtimes are distributed. The notebook output does not provide summary statistics such as mean, median, or exact modal duration, so no precise runtime conclusion is reported here. The source code's plt.show is written without parentheses in this cell, which may prevent the plot from being explicitly displayed when the cell is run.

## 6.8 Release year and movie duration

A scatter plot compares release_year with the numeric movie duration. This can reveal broad patterns, clusters, and unusual values, but the notebook does not print a correlation coefficient or model result. Therefore, the plot should be treated as exploratory rather than evidence of a quantified relationship.

## 6.9 Directors and cast members

The notebook counts director values to display the ten directors with the most listed titles. It also splits the cast field into individual names and counts their appearances. The charts are generated, but exact names and counts are not included in the captured text output, so no numerical ranking is reproduced in this report.

## 6.10 Time gap between release and addition

The notebook subtracts release_year from year_added and plots the non-negative differences in a histogram. This estimates the gap in calendar years between a title's release year and its addition year. It is not an exact elapsed-time calculation because it does not use the full release date, and the chart output does not include a median or other summary statistic.

### 6.11 Monthly additions and rating/genre comparisons

The notebook extracts month names from date_added and plots counts in calendar order. It also uses a cross-tabulation to compare ratings for Movies and TV Shows and a stacked horizontal bar chart to compare content types within popular genres. These visualisations provide useful comparisons, but the notebook does not print the underlying monthly counts or full cross-tabulation in the captured output.

## 7. Key Findings

- The dataset contains 8,807 title records: 6,131 Movies and 2,676 TV Shows.
- Movies account for approximately 69.6% of records, while TV Shows account for approximately 30.4%.
- The highest release-year count shown is 2018, with 1,147 titles.
- TV-MA (3,207) and TV-14 (2,160) are the two most frequent ratings in the printed results.
- The notebook investigates time-based additions, country and genre representation, duration, directors, cast, and content-type comparisons through visualisations.
- Several charts are present without printed numerical summaries; conclusions about those charts should be confirmed by inspecting or exporting the plotted values.

## 8. Limitations

- The analysis depends on the records and metadata in the supplied CSV snapshot; it should not be treated as a live Netflix catalogue.
- Some fields contain missing values, and the effect of missingness may differ across analyses.
- Country, cast, and genre fields can contain multiple values per title; exploding these fields means counts may exceed the number of unique titles.
- The notebook does not consistently print the numerical results behind every chart.
- A few plotting cells use plt.show without parentheses, and one chart title appears to contain a formatting typo; these should be corrected before final presentation.
- The analysis is descriptive. It does not establish why catalogue patterns occur, how viewers behave, or whether a title is successful.

## 9. Recommendations for Improvement

- Add a data-quality section that reports missing values and duplicate records.
- Print or export the top-country, genre, director, actor, and monthly-count tables alongside the charts.
- Calculate summary statistics such as median movie duration and median release-to-addition gap.
- Use consistent date parsing with errors='coerce' and check how many dates fail to parse.
- Correct plt.show calls to plt.show() and review chart titles, labels, and spelling.
- Save the final charts as image files and use consistent colours, figure sizes, and readable axis labels.
- Document the source and date of the dataset so readers understand its coverage.

## 10. Conclusion

This project demonstrates how Python can be used to explore a streaming-content catalogue through data cleaning, aggregation, and visualisation. The visible results show that Movies outnumber TV Shows in the dataset, 2018 is the leading release year in the printed top-ten table, and TV-MA and TV-14 are the most common ratings in the rating counts. The notebook also establishes a useful framework for examining countries, genres, additions over time, runtimes, directors, cast members, and content-type differences. Printing additional summary tables and fixing minor plotting issues would make the analysis more complete and easier to reproduce.

## Appendix: Main Python workflow

The notebook's main workflow is: import pandas and plotting libraries; read netflix_titles.csv; prepare date fields; group and count relevant columns; split multi-value fields where necessary; and generate charts to communicate the results. The report intentionally avoids inventing numerical results for charts whose values were not printed in the notebook.